

Lab 01 — Nmap Reconnaissance

Network Port Discovery & Service Detection

SOC Analyst Level Documentation

Saravanan | github.com/Saravanan-Patrick | March 2026

■■ This is a fully isolated, controlled lab environment. No real systems were targeted. All simulations performed on personal VMs for educational purposes only.

What is Nmap? (Simple Explanation)

■ Think of Nmap like a security guard doing rounds in a building — knocking on every door (port) to check which ones are open, locked or don't exist. It tells you exactly which services are running behind each open door.

Nmap (Network Mapper) is a free and open source tool used by security professionals worldwide to discover devices on a network and find out which services (ports) they are running. In cybersecurity, attackers use Nmap first to understand what they are dealing with before launching an attack. SOC analysts use Nmap to understand what an attacker can see from outside.

Term	Simple Meaning	Real World Example
Port	A virtual door on a computer	Door number 445 = File sharing room
Open Port	Door is open — service is running	Anyone can knock and get a response
Closed Port	Door exists but is locked	Port exists but no service behind it
Filtered Port	Door is hidden by firewall	Firewall blocks — no response at all
Service	What is running behind the open door	Port 445 runs SMB file sharing
Version	Which version of service is running	SMB version 2.0 or 3.0

Lab Environment

Component	Details	Role
Hypervisor	VMware Workstation Pro 25H2	Hosts both virtual machines
Attacker Machine	Kali Linux 2025.4 (amd64)	Runs Nmap — performs the scan
Defender Machine	Windows 10 x64	Target — being scanned
Network Type	VMware LAN Segment (testpractice)	Isolated — no internet
Attacker IP	192.168.20.11 (Static)	Kali Linux IP address
Defender IP	192.168.20.10 (Static)	Windows 10 IP address
Tool Used	Nmap Version 7.95	Network scanner
Goal	Discover open ports on Win10	Find attack surface

SECTION 1 — What To Do First (Pre-Requisites)

Before running Nmap you need to prepare both machines correctly. Skipping any of these steps will cause the scan to fail or give incorrect results.

1

Boot Both VMs in VMware

Open VMware Workstation → Start Kali Linux VM → Start Windows 10 VM → Wait for both to fully load

2

Verify Both VMs Are on Same LAN Segment

In VMware → Right click each VM → Settings → Network Adapter → Confirm both show LAN Segment: testpractice

3

Disable Windows 10 Firewall

■■ IMPORTANT: Windows Firewall must be OFF before scanning. If firewall is ON all ports will show as FILTERED and Nmap cannot detect any services.

On Windows 10 — open CMD as Administrator and run:

```
netsh advfirewall set allprofiles state off
```

■ Expected Output

```
Ok.

# This means firewall is now disabled on all profiles

# (Domain, Private and Public)
```

4

Verify Connectivity — Ping Test from Kali

On Kali terminal — confirm Kali can reach Win10:

```
ping 192.168.20.10
```

■ Expected Output

```
PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.

64 bytes from 192.168.20.10: icmp_seq=1 ttl=128 time=0.5 ms

64 bytes from 192.168.20.10: icmp_seq=2 ttl=128 time=0.4 ms

4 packets transmitted, 4 received, 0% packet loss
```

■ Real Output (Received from Lab)

```
PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.  
  
64 bytes from 192.168.20.10: icmp_seq=1 ttl=128 time=0.45 ms  
  
4 packets transmitted, 4 received, 0% packet loss, time 196013ms  
  
# Confirmed — both machines can communicate!
```

SECTION 2 — Nmap Parameters Explained

Understanding Nmap Flags

Nmap uses flags (parameters) to control what type of scan to perform. Think of these like settings on a camera — different settings give you different kinds of photos. Here are all the flags used in this lab:

Flag	Full Name	What It Does	Simple Meaning
-sV	Service Version	Detects what service and version is running on which open ports	What's open in the open door?
-sS	SYN Scan	Sends SYN packet — fastest and stealthiest scan type	Knock quietly without fully entering
-sn	Ping Scan	Only checks which hosts are alive — no port scanning	See if anyone is home
-sU	UDP Scan	Scans UDP ports instead of TCP	Check the back doors too
-A	Aggressive	Combines -sV -sC -O — full detailed scan	Full inspection of everything
-O	OS Detection	Tries to identify the operating system of the target	What kind of computer is this?
-p	Port Specify	Scan specific ports only instead of top 1000	Only check specific doors
-p-	All Ports	Scan all 65535 ports	Check every single door
-oN	Output Normal	Save results to a text file	Write it all down in a notebook
-oX	Output XML	Save results to XML file for tools	Save in computer readable format
-v	Verbose	Show more details during scan	Give me more information as you go
--top-ports 20	Top Ports	Scan only the most commonly used 20 ports	Just check the most popular doors

SECTION 3 — All Nmap Commands Used in Lab 01

Command 1 — Basic Ping Check

Before scanning we first confirm the target is reachable on the network.

Run on Kali Linux Terminal

```
ping 192.168.20.10
```

Parameter	Value	Meaning
Target IP	192.168.20.10	Windows 10 VM — the defender machine

■ Expected Output

```
PING 192.168.20.10: 64 bytes from 192.168.20.10  
4 packets transmitted, 4 received, 0% packet loss
```

■ Real Output (Received from Lab)

```
173 packets transmitted, 0 received, 100% packet loss  
time 196013ms  
  
# NOTE: First ping failed because Win10 firewall was ON  
  
# After disabling firewall – ping worked successfully
```

Command 2 — Service Version Scan (Main Scan)

This is the primary Nmap command used to discover open ports and identify what services are running.

Run on Kali Linux Terminal

```
nmap -sV 192.168.20.10
```

Parameter	Value	Meaning
nmap	Tool name	Call the Nmap program
-sV	Flag	Detect service versions on each open port
192.168.20.10	Target IP	Scan the Windows 10 defender machine

■ Expected Output

```
Starting Nmap X.XX at YYYY-MM-DD

Nmap scan report for 192.168.20.10

Host is up (0.0005s latency).

PORT      STATE SERVICE VERSION
135/tcp    open  msrpc  Microsoft Windows RPC
139/tcp    open  netbios-ssn Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Microsoft Windows SMB
MAC Address: XX:XX:XX:XX:XX:XX (VMware)

Service Info: OS: Windows

Nmap done: 1 IP address (1 host up) scanned
```

■ Real Output (Received from Lab)

```
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-14 06:48 EDT

Nmap scan report for 192.168.20.10

Host is up (0.00077s latency).

Not shown: 997 closed tcp ports (reset)

PORT      STATE SERVICE VERSION
135/tcp    open  msrpc  Microsoft Windows RPC
139/tcp    open  netbios-ssn Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
MAC Address: 00:0C:29:FD:99:7A (VMware)

Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Nmap done: 1 IP address (1 host up) scanned in 22.51 seconds
```

Command 3 — Save Scan Results to File

This command runs the same scan but saves the output to a text file for documentation and GitHub upload.

Run on Kali Linux Terminal

```
nmap -sV 192.168.20.10 -oN nmap-scan-results.txt
```

Parameter	Value	Meaning
nmap	Tool name	Call the Nmap program
-sV	Flag	Detect service versions
192.168.20.10	Target IP	Windows 10 defender machine
-oN	Output flag	Save output in normal text format
nmap-scan-results.txt	Filename	Name of the file to save results

■ Expected Output

```
# File created: nmap-scan-results.txt  
  
# Contents match terminal output  
  
# File saved to current directory
```

■ Real Output (Received from Lab)

```
# File successfully created on Kali Desktop  
  
# nmap-scan-results.txt confirmed with cat command:  
  
# Nmap 7.95 scan initiated 2026-03-14 06:41 EDT  
  
# Nmap scan report for 192.168.20.10  
  
# Host is up (0.00055s latency).  
  
# PORT STATE SERVICE  
  
# 135/tcp open msrpc  
  
# 139/tcp open netbios-ssn  
  
# 445/tcp open microsoft-ds
```

Command 4 — Aggressive Full Scan

This advanced scan combines service detection, OS detection and default scripts for maximum information gathering.

Run on Kali Linux Terminal

```
nmap -A -sV 192.168.20.10
```

Parameter	Value	Meaning
nmap	Tool name	Call the Nmap program
-A	Flag	Aggressive scan — combines OS detect + scripts + traceroute
-sV	Flag	Detect service versions
192.168.20.10	Target IP	Windows 10 defender machine

■ Expected Output

```
PORT STATE SERVICE VERSION
```

```
135/tcp open msrpc Microsoft Windows RPC
```

```
139/tcp open netbios-ssn Microsoft Windows netbios-ssn
```

```
445/tcp open microsoft-ds Windows 10 SMB
```

```
OS details: Microsoft Windows 10
```

```
Network Distance: 1 hop
```

■ Real Output (Received from Lab)

```
Starting Nmap 7.95 at 2026-03-14 06:48 EDT
```

```
PORT STATE SERVICE VERSION
```

```
135/tcp open msrpc Microsoft Windows RPC
```

```
139/tcp open netbios-ssn Microsoft Windows netbios-ssn
```

```
445/tcp open microsoft-ds?
```

```
MAC Address: 00:0C:29:FD:99:7A (VMware)
```

```
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows
```

```
Nmap done: 1 IP address scanned in 35.89 seconds
```


SECTION 4 — Scan Findings & Analysis

Open Ports Discovered

Port	State	Service	Version	Risk Level	Why It Matters
135/tcp	OPEN	msrpc	Microsoft Windows RPC	MEDIUM	Used for lateral movement in attacks
139/tcp	OPEN	netbios-ssn	Microsoft Windows NetBIOS	MEDIUM	Windows network discovery service
445/tcp	OPEN	microsoft-ds	SMB — Windows File Sharing	HIGH	WannaCry ransomware used this port!
997 ports	CLOSED	—	TCP RST response received	LOW	Ports exist but nothing running

What These Ports Mean — For Non-IT People

Port 135 (RPC): Like a receptionist in an office building — it helps different programs talk to each other remotely. Attackers use this to move between computers inside a company network.

Port 139 (NetBIOS): Like a company directory — it helps Windows computers find and talk to each other on the same network. Old and often unnecessary in modern networks.

Port 445 (SMB): Like a shared filing cabinet — allows Windows computers to share files and printers. This is the MOST dangerous open port. The WannaCry ransomware attack in 2017 infected 200,000 computers worldwide using this exact port!

OS Detection Results

Field	Value
Operating System Detected	Microsoft Windows
CPE (Common Platform Enumeration)	cpe:/o:microsoft:windows
MAC Address	00:0C:29:FD:99:7A
MAC Vendor	VMware Inc
Host Status	Up (0.00077s latency)
Scan Duration	22.51 seconds

SECTION 5 — Indicators of Compromise (IOCs)

IOCs are pieces of evidence that show an attack happened or is happening. A SOC analyst documents these to investigate and respond to incidents.

IOC Type	Value	Significance
Source IP (Attacker)	192.168.20.11	Kali Linux — machine performing the scan
Target IP (Defender)	192.168.20.10	Windows 10 — machine being scanned
Scan Type	TCP SYN Service Scan (-sV)	Attacker fingerprinting the target
Ports Targeted	Top 1000 TCP ports	Automated reconnaissance activity
Open Ports Found	135, 139, 445	Attack surface identified
Tool Identified	Nmap 7.95	Visible in Wireshark packet headers
Date and Time	2026-03-14 06:39 EDT	Attack timestamp
Scan Duration	22-35 seconds	Automated — not manual
OS Identified	Microsoft Windows	Attacker knows OS — targeted exploit possible
MAC Address	00:0C:29:FD:99:7A	VMware virtual machine confirmed
Packets Generated	2,248 packets	High volume — SIEM would alert

SECTION 6 — Wireshark Traffic Analysis

While Nmap was running, Wireshark captured all network packets to document exactly what the attack traffic looks like from a defender perspective.

Wireshark Filter Used	Purpose	What It Showed
ip.addr == 192.168.20.10	All traffic to Win10	Complete attack traffic visible
tcp.flags.syn == 1 and tcp.flags.ack == 0	SYN packets only	Port scan pattern — 1000 SYN packets
icmp	Ping traffic	Network discovery pings
tcp.port == 445	SMB traffic	Netcat C2 connection attempt
udp	UDP traffic	UDP port scan packets

Capture Metric	Value
Total Packets Captured	2,248 packets
Packets Dropped	0 (0.0%)
Protocols Seen	TCP, NBSS, SMB, Microsoft Windows Browser Protocol
Key Finding	TCP SYN packets sent to all 1000 ports — classic port scan signature

SECTION 7 — MITRE ATT&CK; & Cyber Kill Chain Mapping

Kill Chain Stage	MITRE ATT&CK ID	Technique	What Happened in Lab
Reconnaissance	T1046	Network Service Discovery	Nmap scanned top 1000 TCP ports on Win10
Reconnaissance	T1595	Active Scanning	SYN packets sent to all ports
Reconnaissance	T1592	Gather Host Information	OS and service versions identified

SECTION 8 — Security Observations & Recommendations

■ Port 445 SMB is Extremely Dangerous

Port 445 running SMB (Server Message Block) is the most dangerous finding in this scan. SMB port 445 was the exact vulnerability exploited by WannaCry ransomware in 2017 which infected over 200,000 computers across 150 countries causing billions in damages. In production environments this port should NEVER be exposed to untrusted networks.

■ Windows Firewall is Effective

When Windows Firewall was enabled all 1000 scanned ports appeared as FILTERED — meaning Nmap received no response and could not identify any services. This demonstrates that a simple host-based firewall is extremely effective at hiding the attack surface from reconnaissance scans.

■ Nmap Scan is Detectable

The Nmap scan generated 2,248 packets in approximately 22 seconds — scanning 1000 ports in rapid succession. Any SIEM monitoring this network would immediately trigger a port scan alert based on the pattern of rapid sequential TCP SYN packets from a single source IP.

■ MAC Address Reveals VM Environment

The MAC address 00:0C:29:FD:99:7A clearly identifies the target as a VMware virtual machine. In real attacks this information helps attackers understand they may be in a sandboxed or lab environment.

■ OS Detection Enables Targeted Attacks

By detecting the OS as Microsoft Windows, an attacker now knows exactly which exploits to use. For example knowing the target runs Windows and has port 445 open directly points to EternalBlue exploit — the same one used in WannaCry.

SECTION 9 — What I Learned

- How to perform network reconnaissance using Nmap from Kali Linux
 - How Nmap flags work and when to use each one (-sV, -A, -sS, -oN)
 - How open ports reveal a computer's attack surface to an attacker
 - Why port 445 SMB is critically dangerous and its connection to WannaCry ransomware
 - How Windows Firewall completely hides open ports from Nmap scans
 - How to capture and analyze attack traffic in Wireshark alongside Nmap
 - How to identify port scan patterns from TCP SYN packets in Wireshark
 - How to document findings as Indicators of Compromise (IOCs)
 - How Nmap reconnaissance maps to MITRE ATT&CK; technique T1046
 - How attackers use reconnaissance to plan their next attack step
-

SECTION 10 — Errors Encountered & Fixes Applied

The following errors were encountered during Lab 01. Each error is documented with what happened and the exact fix that was applied:

■■ ERROR: First Nmap Scan — All 1000 Ports Showed as FILTERED

What Happened: When Nmap was run for the first time the output showed: Not shown: 1000 filtered tcp ports (no-response). Zero open ports were detected and no services were identified. This happened because Windows Defender Firewall was still enabled on Windows 10, blocking all incoming connection attempts from Kali.

■ Fix Applied: Disabled Windows Defender Firewall on Win10 via CMD as Administrator: `netsh advfirewall set allprofiles state off` — After disabling, re-ran `nmap -sV 192.168.20.10` and immediately found 3 open ports: 135, 139 and 445.

■■ ERROR: First Ping — 100% Packet Loss (173 packets, 0 received)

What Happened: When the first ping was run from Kali to Win10 the result showed 173 packets transmitted, 0 received, 100% packet loss. The machines could not communicate at all. This happened because both VMs were not on the same LAN Segment, OR Windows Firewall was blocking ICMP ping requests.

■ Fix Applied: Confirmed both VMs were assigned to the same LAN Segment (testpractice) in VMware settings AND disabled Windows Firewall. After both fixes ping worked successfully with 0% packet loss.

■■ ERROR: Nmap -sV Showed microsoft-ds? with Question Mark

What Happened: The Nmap service version scan showed port 445 as microsoft-ds? with a question mark. This means Nmap detected the port was open but could not fully identify the exact version of the service running on port 445.

■ Fix Applied: This is normal behaviour for Windows 10 SMB — Microsoft has added protections that prevent full version fingerprinting. The port is confirmed open and identified as SMB. The question mark simply means version fingerprinting was incomplete — not an error.

Lab 01 — Final Summary

Item	Result
Lab Status	Complete
Target Machine	Windows 10 (192.168.20.10)

Attacker Machine	Kali Linux (192.168.20.11)
Tool Used	Nmap 7.95
Open Ports Found	3 ports — 135, 139, 445
Most Critical Port	445/tcp — SMB (High Risk)
OS Detected	Microsoft Windows
Packets Captured	2,248 packets in Wireshark
Files Saved	nmap-scan-results.txt + wireshark_eth05WIAM3.pcapng
MITRE Technique	T1046 — Network Service Discovery
Kill Chain Stage	Reconnaissance
Next Step	Lab 02 — Brute Force Attack on port 445

Lab 01 Complete! ■ Port 445 discovered — Lab 02 brute force next!

github.com/Saravanan-Patrick/Lab-01-Nmap-Reconnaissance